WGAN-GP & Discriminator Optimal Transport

Improved Training of Wasserstein GANs & DOT Sampling for MNIST Generation

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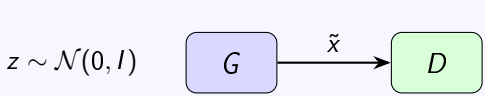
Why GANs Need Better Training

Classical GAN



Min-max game:

$$\min_G \max_D \mathbb{E}_{x \sim p_r}[\log D(x)] + \mathbb{E}_{z \sim p_z}[\log(1 - D(G(z)))]$$

Three Major Problems

1. **Vanishing gradients**
   If D is too good, G gets no useful signal
2. **Mode collapse**
   G produces limited variety
3. **Training instability**
   Sensitive to hyperparameters, oscillations

Wasserstein Distance

JS Divergence: The Problem

When $\text{supp}(p_r) \cap \text{supp}(p_g) = \emptyset$:

$$\text{JS}(p_r, p_g) = \log 2 \quad \text{(constant!)}$$

$\Rightarrow$ No gradient for G to follow.

Wasserstein-1: The Solution

- Smooth even when supports are disjoint
- Provides meaningful gradients everywhere
- Correlates with sample quality

Kantorovich–Rubinstein Duality

Eq. (1)

$$\boxed{W_1(p_r, p_g) = \sup_{\|f\|_{\text{Lip}} \leq 1} \Big[\mathbb{E}_{x \sim p_r}[f(x)] - \mathbb{E}_{y \sim p_g}[f(y)] \Big]}$$

Key constraint: the critic D must be **1-Lipschitz**, i.e. $\|D\|_{\text{Lip}} \leq 1$.

Weight Clipping: Why WGAN Fails

WGAN (Arjovsky, 2017) clips weights in $[-c, c]$ to enforce Lipschitz.

Three Failure Modes

1. **Capacity underuse**
   Critic learns overly simple functions
2. **Exploding/vanishing gradients**
   Gradient norms diverge across layers
3. **Bimodal weights**
   Weights pushed to $\pm c$ boundaries

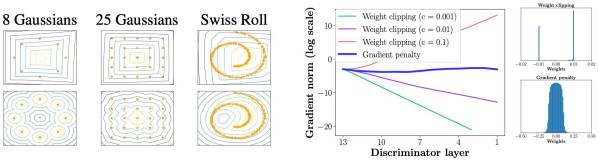


Figure 1 (Gulrajani et al., 2017): Value surfaces, gradient norms, weight distributions.

Gradient Penalty: The WGAN-GP Solution

Proposition 1 (Gulrajani et al., 2017)

Let f^* be the optimal 1-Lipschitz critic and π the optimal coupling between p_r and p_g. For $x_t = t\,x + (1-t)\,y$ with $(x, y) \sim \pi$ and $0 \le t \le 1$:

$$\mathbb{P}_{(x,y)\sim\pi}\left[\nabla f^*(x_t) = \frac{y - x_t}{\|y - x_t\|}\right] = 1$$

$\Rightarrow$ Optimal critic has gradient norm 1 a.e. along interpolations between paired real/fake samples.

Idea: Penalize when $\|\nabla_{\hat{x}} D(\hat{x})\|_2 \neq 1$ on interpolated points:

$$\hat{x} = \varepsilon x + (1-\varepsilon)\tilde{x}, \qquad \varepsilon \sim \mathrm{Uniform}[0,1]$$

Critic Loss with Gradient Penalty *Eq. (2)*

$$\boxed{\mathcal{L}_D = \underbrace{\mathbb{E}_{\tilde{x}\sim p_g}[D(\tilde{x})] - \mathbb{E}_{x\sim p_r}[D(x)]}_{\text{Wasserstein loss}} + \underbrace{\lambda\,\mathbb{E}_{\hat{x}}\left[\left(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1\right)^2\right]}_{\text{Gradient Penalty } (\lambda=10)}}$$

Layer Normalization (No Batch Norm!)

Why Batch Norm is Forbidden

- BN correlates examples within a batch
- GP requires **per-example** gradients
- BN invalidates the GP computation

Section 4: "No critic batch normalization"

Layer Normalization

- Normalizes **per example**
- μ, σ^2 computed over features of each sample
- No inter-example dependency

Layer Norm Formula

Eq. (3)

$$\boxed{\mathrm{LN}(h) = \gamma \odot \frac{h - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta}$$

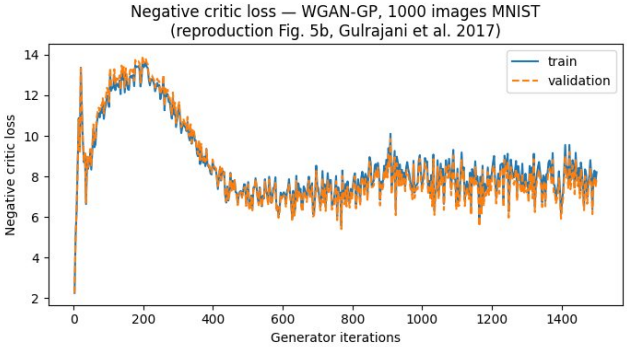
where μ, σ^2 computed **per example**

Our Critic Architecture (C3)

WGANCritic v2 Design Choices

- ✓ **Spectral Normalization**
  Guarantees $\|D\|_{\text{Lip}} \leq 1$
- ✓ **Layer Normalization**
  No batch dependency
- ✓ **Dropout** ($p = 0.1$)
  Light regularization
- × **No Batch Norm**
  Would invalidate GP

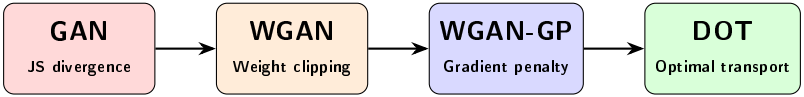
Training: $n_{\text{critic}} = 5$ critic updates per generator step (5:1 ratio).



Negative critic loss — WGAN-GP, 1000 images MNIST.
Reproduction of Fig. 5b (Gulrajani et al., 2017).

Result: No overfitting — train and validation loss overlap even on 1000 images.

Context: From WGAN-GP to DOT



Key Insight

A trained WGAN critic approximates the **dual potential** of optimal transport.
DOT uses this connection to **improve generated samples after training**.

Kantorovich Duality: Primal & Dual

Primal (Optimal Transport)

$$W_1(p_r, p_g) = \min_{\pi \in \Pi(p_r, p_g)} \mathbb{E}_{(x,y)\sim\pi}\big[\|x - y\|_2\big]$$

Find the cheapest transport plan π.

Dual (Kantorovich–Rubinstein)

$$W_1(p_r, p_g) = \max_{\|f\|_{\mathrm{Lip}} \le 1} \Big[\mathbb{E}_{x\sim p_r}[f(x)] - \mathbb{E}_{y\sim p_g}[f(y)]\Big]$$

f^* = optimal dual potential.

Key Link

$$f^* \text{ (optimal dual potential)} \longleftrightarrow \text{Transport map } T$$

The trained critic D approximates $f^* \Rightarrow D$ encodes the optimal transport direction.

Spectral Normalization for Lipschitz Constraint

Spectral Normalization (Miyato et al., 2018) *Eq. (4)*

$$\boxed{W_{\mathrm{SN}} = \frac{W}{\sigma(W)}} \qquad \sigma(W) = \text{largest singular value of } W$$

$$\Rightarrow \text{Guarantees } \|D\|_{\mathsf{Lip}} \leq 1 \text{ globally (layer by layer).}$$

Why SN matters for DOT:

- $\|D\|_{\mathsf{Lip}} \leq 1 \Rightarrow D$ is a valid candidate for the dual problem
- Lipschitz constant stays controlled $\Rightarrow$ stable k_{eff} estimation
- Without SN, DOT performance degrades significantly

DOT: Active Sample Improvement

Rejection Sampling (Passive)

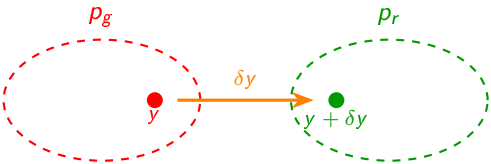
Generate $\rightarrow$ reject bad samples

Wastes compute, doesn't improve G

DOT (Active)

Generate $\rightarrow$ **move** samples toward p_r

Uses D to find transport direction



$$D(y + \delta y) > D(y)$$

δy follows the **optimal transport direction**.

Theorem 1: D Encodes Optimal Transport

Theorem 1 (Tanaka, 2019)

If $\tilde{D} = D/K$ is the normalized optimal critic, then the optimal transport map is:

$$\boxed{T(y) = \arg\min_x \left\{ \|x - y\|_2 - \tilde{D}(x) \right\}}$$

Interpretation:

- $T(y)$ maps each generated sample y to its nearest real counterpart
- $\tilde{D} = D/K$ is the normalized dual potential
- Formal bridge between GAN training and optimal transport theory

Problem: K (Lipschitz constant of D) is **unknown** in practice.

In Practice: Estimating k_{eff}

Lower Bound on W_1 *Eq. (5)*

$$V_D(G, D) \leq K \cdot W_1(p_r, p_g)$$

Maximizing $V_D \Rightarrow D$ approximates the dual potential f^*.

Problem: K is unknown

- $K = \|D\|_{\text{Lip}}$ cannot be computed exactly
- Needed to normalize: $\tilde{D} = D/K$

Solution: Estimate k_{eff}

- Empirical estimation from random pairs
- SN keeps k_{eff} stable during training
- Paper: use enough samples

Effective Transport Formula

Target-Space DOT *Eq. (22), Tanaka 2019*

$$T_D^{\text{eff}}(y) = \arg\min_x \left\{ \|x - y\|_2 - \frac{1}{k_{\text{eff}}} D(x) \right\}$$

Interpretation: For each generated image $y = G(z)$, find x that:

1. Stays close to y ($\|x - y\|$ term)
2. Maximizes critic score ($-D(x)/k_{\text{eff}}$ term)

$\Rightarrow$ Moves y toward a more realistic region.

✓ Works in **low dimension**

× Creates adversarial artifacts in **high dimension** (images)

Latent-Space DOT: Our Method

Latent-Space DOT *Eq. (6) / Eq. (23), Tanaka 2019*

$$\boxed{T^{\text{eff}}_{D\circ G}(z_y) = \arg\min_z \left\{ \|z - z_y\|_2 - \frac{1}{k_{\text{eff}}} D(G(z)) \right\}}$$

Why latent space?

- $z \in \mathbb{R}^{100}$ vs images $\in \mathbb{R}^{784}$ — lower dimension, more stable
- G constrains outputs to valid images — no adversarial artifacts

Our Improvements (v2)

- **LR warm-up** (5 first steps at $\mathrm{lr}/5$) — avoids large initial jumps
- **Clipping** $z \in [-4, 4]$ — stays in Gaussian support
- **Percentile 95** for k_{eff} instead of max — more robust to outliers

Algorithm 2: Latent-Space DOT

Algorithm 2: Latent-space DOT with our v2 modifications

Input: G, D (trained), k_{eff}, sample z_y, lr ε, offset δ

$z \leftarrow z_y$
for $\text{step} = 1$ **to** n_{updates} **do**
  $g \leftarrow \nabla_z \left\{ \|z - z_y + \delta\|^2 - \frac{1}{k_{\text{eff}}} D(G(z)) \right\}$
  $g \leftarrow g - \frac{(g \cdot z)}{\sqrt{d}}\, z$ // spherical projection
  $z \leftarrow z - \varepsilon \cdot g$ // Adam ($\beta_1 = 0, \beta_2 = 0.9$)
  $z \leftarrow \text{clip}(z, [-4, 4])$ // our improvement
end for
return $x = G(z)$

Hyperparameters: $n_{\text{updates}} = 50$, $\varepsilon = 0.01$, $\delta = 10^{-3}$, warm-up = 5 steps.

Architecture Comparison: C1 → C2 → C3

Critic	SN	LN	Dropout	FID ↓	Prec. ↑	Recall ↑
C1 (baseline)	×	×	×	≈385	0.00	0.00
C2 (v1, +LN)	×	✓	×	≈43	0.46	0.18
C3 (v2, +SN)	✓	✓	✓	**≈20**	**0.62**	**0.27**

Key Finding

Spectral Normalization is necessary (not optional) for DOT.
Without SN (C2), k_{eff} is erratic and DOT is unreliable. SN alone drops FID from ≈43 to ≈20.

Sampling Methods Comparison

Method	FID ↓	Precision ↑	Recall ↑
Baseline ($z \sim \mathcal{N}(0, I)$)	≈17.1	0.61	0.31
Hard truncation ($\psi = 1.5$)	≈16.8	0.62	0.30
Soft truncation ($\psi = 0.7$)	≈16.5	0.62	0.29
DOT v2 (ours)	**15.79**	**0.63**	0.28

DOT v2: Best FID

15.79 (vs ≈17.1 baseline)
Precision 0.63 (vs 0.61)

Trade-off

DOT improves **precision** (sharpness) but slightly reduces **recall** (diversity).

Conclusion & Limitations

Main Results:

- ✓ SN is **necessary** for DOT — not optional (FID: 43 $\rightarrow$ 20)
- ✓ DOT v2 achieves best FID: **15.79** with Precision 0.63
- ✓ Gradient penalty + Layer Norm = stable WGAN-GP training

Limitations:

- △ Trade-off: Precision 0.63 vs Recall 0.28
- × MLP generator is fixed (project constraint)

Future Work:

- CNN generator (DCGANs) to improve capacity
- Reduce n_{updates} for faster DOT inference
- Adaptive k_{eff} estimation during sampling

Thank You

Questions?

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Gulrajani et al. (2017) — Improved Training of Wasserstein GANs
Tanaka (2019) — Discriminator Optimal Transport
Miyato et al. (2018) — Spectral Normalization for GANs